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J. E. GORTNER

1,744,074

RADIATOR UNIT

Filed Feb. 18, 1929

Fig. 1.

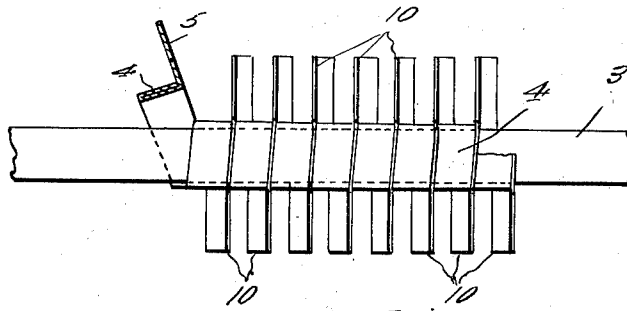


Fig. 2.

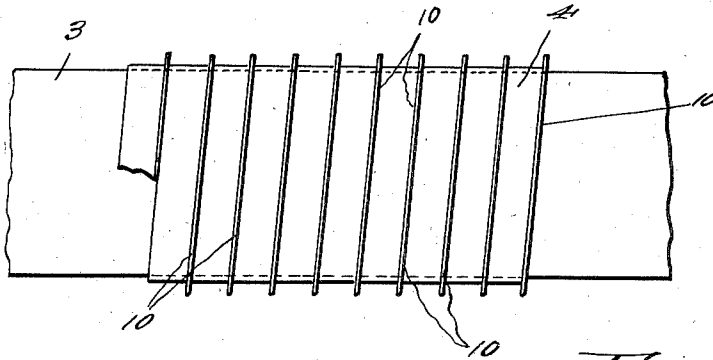


Fig. 4.

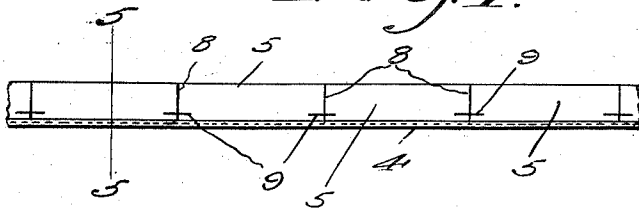


Fig. 3.

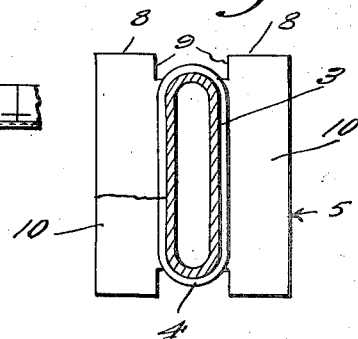


Fig. 5.

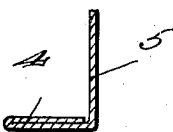
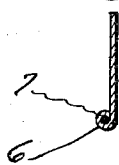


Fig. 6.



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RADIATOR UNIT

REISSUED

Application filed February 18, 1929. Serial No. 340,832.

The present invention relates to new and useful improvements in the art generally known as heat exchange, and more particularly the invention has reference to a novel radiator unit which may be employed in the construction of heat transfer or condensation equipment, cooling systems, radiators for locomotives, gasoline engines, aeroplanes, and condensers for all purposes.

10 The principal object of this invention is to provide a radiator unit including a water tube which is capable of being readily repaired without discarding the water tube proper providing the latter is not ruptured.

15 Another salient object is to provide a radiator unit having a removable fin construction.

Another very important object is to provide a fin construction capable of being readily applied to radiator constructions not originally provided with cooling fins.

20 Still another object is to provide a cooling fin for radiator water tubes which is capable of being manufactured at low cost because of its extremely brief construction.

25 These and other objects of the invention will become more apparent to the reader, after considering the invention as described and claimed hereinafter.

30 In the drawings:—
Figure 1 represents a fragmentary partial sectional view of the improved unit.

Fig. 2 is a fragmentary side elevation of the radiator unit.

35 Fig. 3 is a cross sectional view of the unit.
Fig. 4 is a fragmentary elevation of the fin construction shown in extended position.

Fig. 5 is a cross sectional view of the fin taken substantially on the line 5—5 of Fig. 4.

40 Fig. 6 is a cross sectional view of a slightly modified form of fin.

Referring to the drawings wherein like numerals designate like parts, the improved radiator unit includes a water tube 3, which may be of any cross sectional shape, but for the purpose of illustration, and explanation, the water tube of substantially rectangular cross section has been elected.

50 The novel fin construction embodies an elongated strip of suitable ductile material,

preferably copper, constructed in the following manner for convolution on the water tube 3. The strip has one longitudinal edge portion bent upon itself as at 4 for obviously reinforcing this portion of the strip which must contact with the tube 3.

The opposite edge portion of the strip is disposed laterally to provide a flange 5. In Fig. 6 a slightly modified form of the fin is shown, wherein the strip is rolled at one longitudinal edge portion as at 6 over a reinforcing element 7. The flange construction of each form is provided at predetermined spaced intervals with transversely extending slits 8 extending inwardly from the free edge of the flange. The flange of each form is provided with longitudinally extending and spaced slits 9, each intersecting one of the transverse slits 8.

It will thus be seen that by convoluting the strengthened edge portion of the strip on the water tube 3, the flanges will break at the slit 8 and due to the intersection of the slits 8 by the slits 9, the flange will assume the character of individual wings 10, projecting beyond the opposite sides of the tube 3. Obviously in the manufacture of this novel unit, various changes may be made in the shape, size, materials, without departing from the spirit and scope of the invention as claimed hereinafter.

Having thus described my invention, what I claim as new is:—

A radiator unit comprising a water tube, and a fin, said fin comprising an elongated strip of relatively heavy material and a flange at one longitudinal edge thereof, said flange being slitted transversely from its free edge at predetermined intervals substantially equal to the width of the water tube, said flange being provided with longitudinally extending and spaced slits, each intersecting one of the transverse slits.

In testimony whereof I affix my signature.
JAMES E. GORTNER.